

WEEK 2



Loops

Say it once. Do it many times.

반복문 — 한 번 쓰면 여러 번 실행 · 영어코딩

BIG PICTURE

Stop copy-pasting 🤔



Last week: 5 lines for 5 blocks. This week: 1 loop.

같은 줄을 반복하지 않아도 됩니다

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

The `for` loop

```
for i in range(5):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

`range(5)` = do the inside **5 times**.

`range(5)` = 5번 반복

CONCEPT

Indent = inside the loop

```
for i in range(5):  
    agent.place(DOWN)    # repeats ✓  
    agent.move(FORWARD, 1) # repeats ✓  
agent.turn(RIGHT)      # runs once ✗
```

Indented lines repeat. Un-indented lines do not.

WATCH IT!

The loop runs 3 times



`i = 0`

```
▸ blocks placed
```

CODE

A row vs a tower

```
# flat row on the ground
for i in range(6):
    agent.place(DOWN)
    agent.move(FORWARD, 1)

# tall tower standing up
for i in range(6):
    agent.place(DOWN)
    agent.move(UP, 1)
```


PREDICT 🤖

What does this build?

Predict first — then click reveal

```
for i in range(4):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

Reveal answer 🔍 reveal

VOCAB

New words today

<code>loop</code>	code that repeats a set of commands
<code>for</code>	a loop that repeats a fixed number of times
<code>range(n)</code>	repeat n times (counts 0 to n-1)
<code>indent</code>	the spaces that put a line inside the loop

반복문

~동안

범위

들여쓰기

 DEBUG

What's wrong here?

buggy

```
# goal: a flat row of 6 blocks  
for i in range(6):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)  
    agent.turn(RIGHT)
```

Think first — then click next.



DEBUG

The fix

Bug: `agent.turn(RIGHT)` is inside the loop, so the agent turns after every block and walks in a tiny circle instead of a straight line.

fixed

```
for i in range(6):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)
```

YOUR TURN!

Loop a 20-block wall

- ✓ Use one `for` loop with `range(20)`
- ✓ Place a block + move forward each time
- ✓ Then build a U-shaped wall (3 sides)
- ✓ Send a photo + walkthrough on KakaoTalk



QUIZ

QUESTION 1 OF 3

How many times does `for i in range(5):` repeat?

4 times.

5 times.

Forever.

6 times.

 QUIZ

QUESTION 2 OF 3

What's wrong?

```
for i in range(6):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)  
    agent.turn(RIGHT)
```

DOWN should be UP.

You can't place and move together.

range(6) is too big.

turn(RIGHT) is inside the loop — the agent spins in circles.



QUIZ

QUESTION 3 OF 3

What does the **indent** control?

Nothing — spaces are ignored.

Which lines are inside the loop and repeat.

How fast the agent moves.

The color of the blocks.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Loops unlocked!

Next: variables — name your numbers.

다음 주: 변수로 숫자에 이름 붙이기